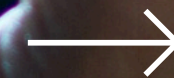


TIDY01: A Short Course in R Basics

Presented by Gabriel Rodrigues Palma



Tidyverse for Ecologists
(TIDY01)



Day 1 (9:30 – 17:30)

R Syntax & Variables
(9:35 – 11:05)

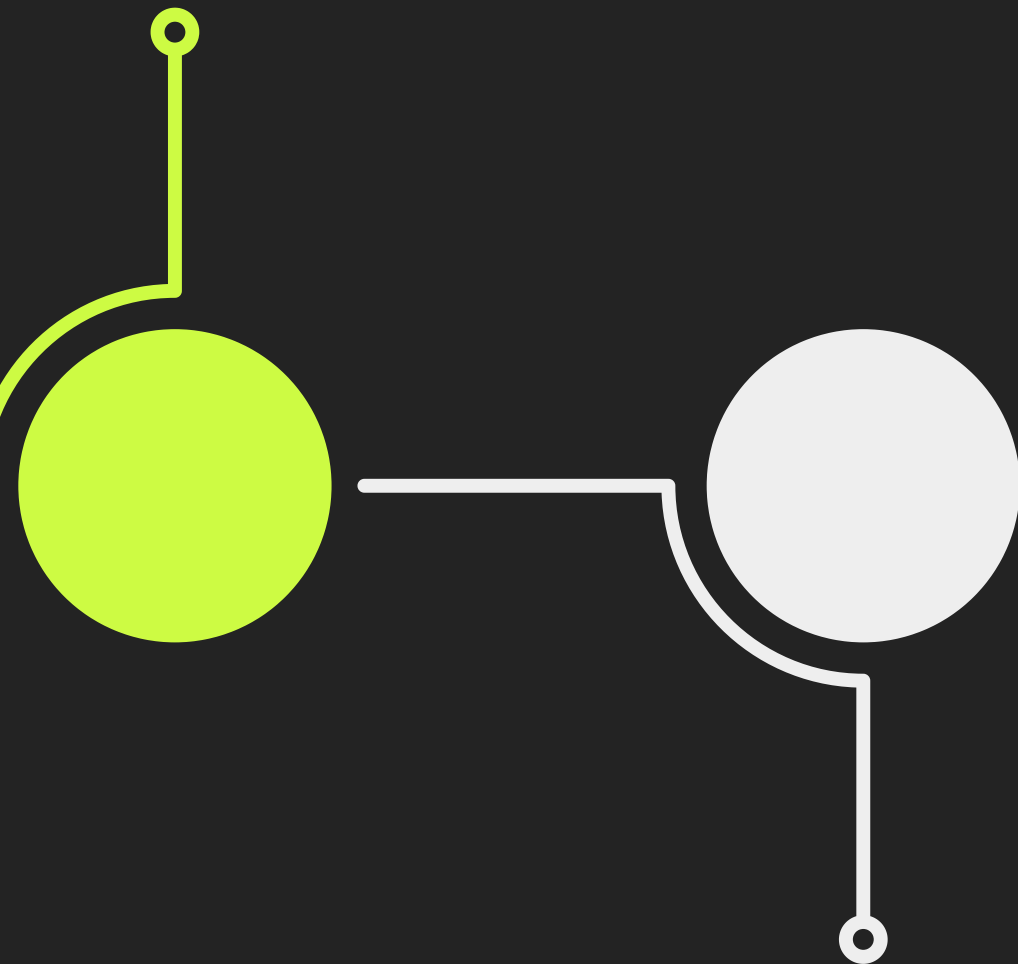


Tidyverse for Ecologists
(TIDY01)



Day 1 (9:30 – 17:30)

R Syntax & Variables
(9:35 – 11:05)



Data Types/Conditionals
(11:05 – 12:30)

Tidyverse for Ecologists
(TIDY01)



Day 1 (9:30 – 17:30)

R Syntax & Variables
(9:35 – 11:05)

Loops & Functions
(13:30 – 15:00)

Data Types/Conditionals
(11:05 – 12:30)

Tidyverse for Ecologists
(TIDY01)



Day 1 (9:30 – 17:30)

R Syntax & Variables
(9:35 – 11:05)

Loops & Functions
(13:30 – 15:00)

Data Types/Conditionals
(11:05 – 12:30)

Data Structures & File
Handling (15:00 – 16:30)

Tidyverse for Ecologists
(TIDY01)



Day 1 (9:30 – 17:30)

R Syntax & Variables
(9:35 – 11:05)

Loops & Functions
(13:30 – 15:00)

Clean Code in R
(16:30 – 17:30)

Data Types/Conditionals
(11:05 – 12:30)

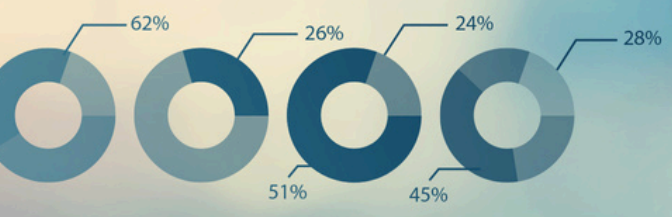
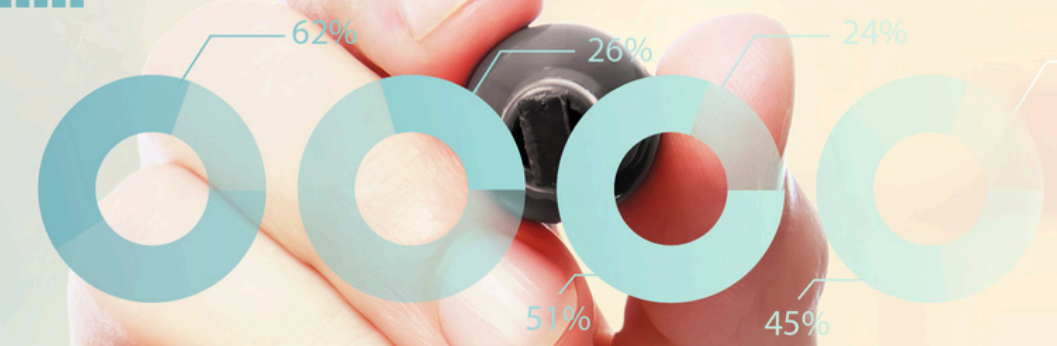
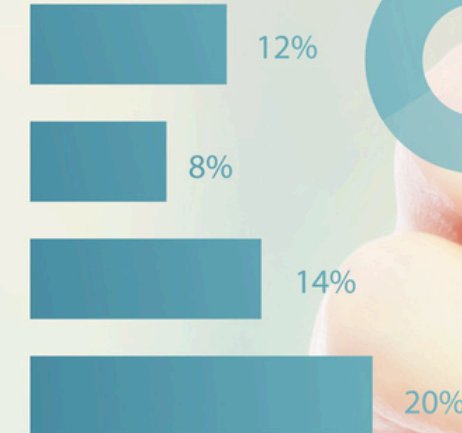
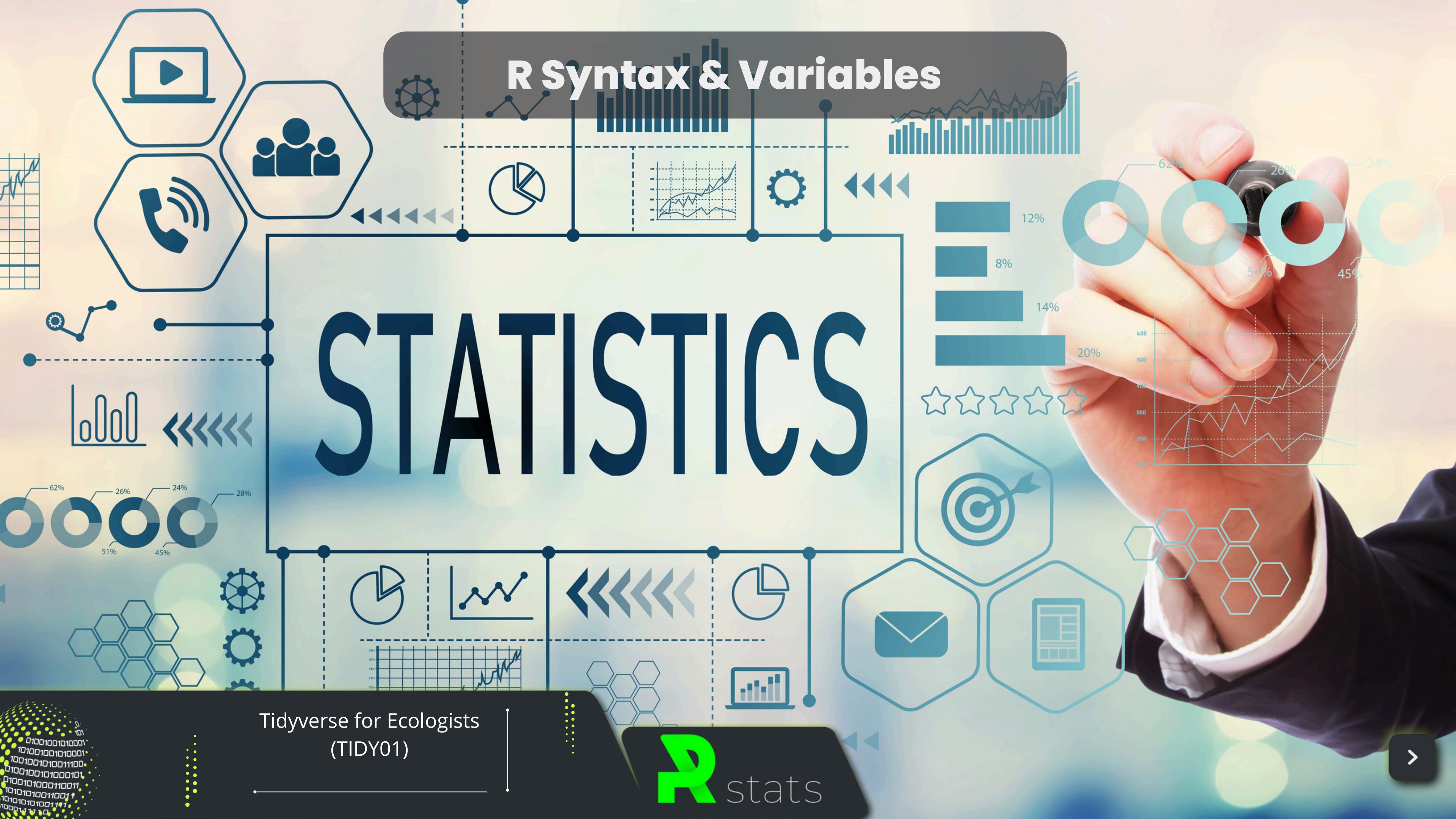
Data Structures & File
Handling (15:00 – 16:30)

Tidyverse for Ecologists
(TIDY01)



R Syntax & Variables

STATISTICS



Tidyverse for Ecologists
(TIDY01)

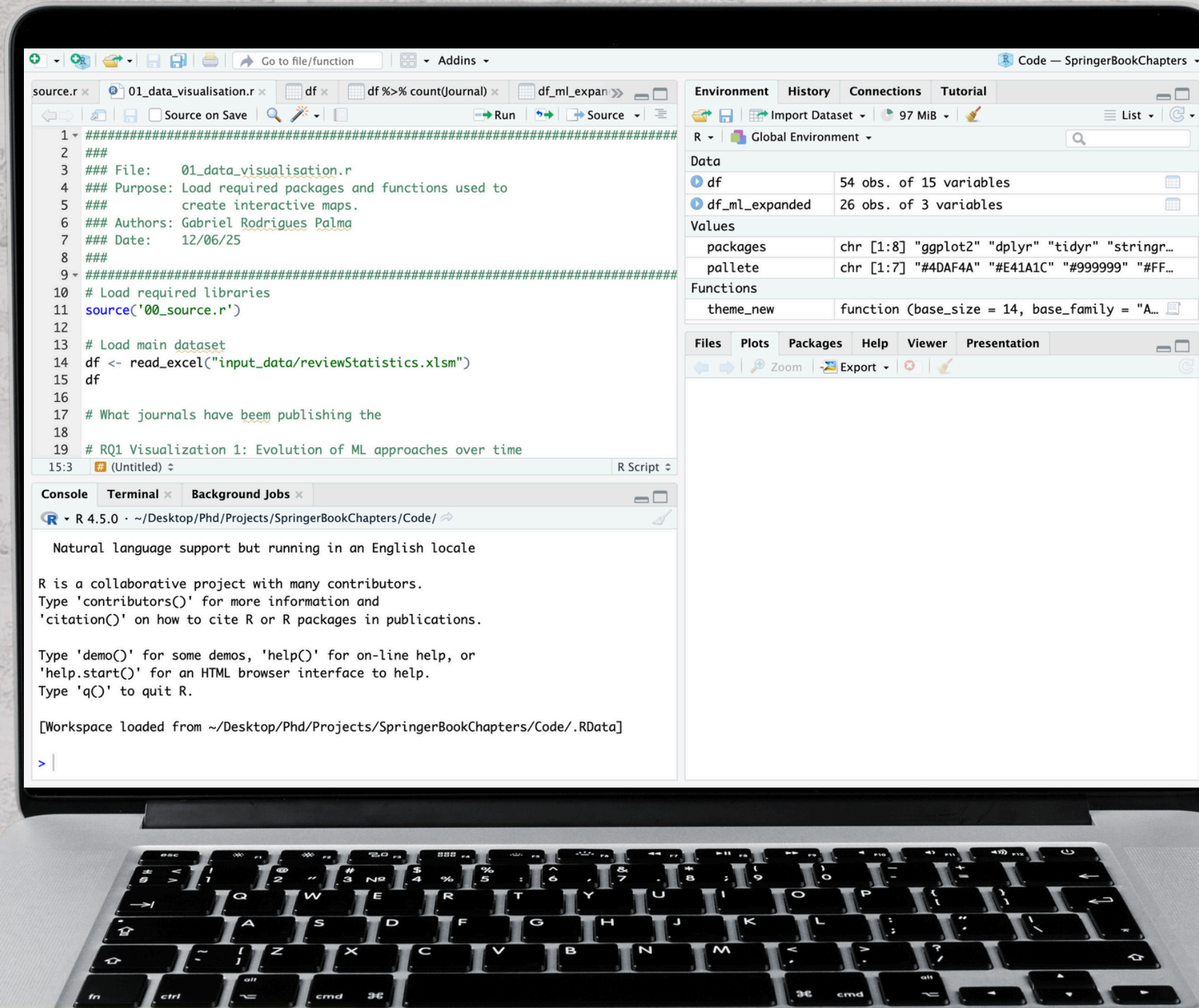


R Syntax & Variables

Tidyverse for Ecologists
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R Syntax & Variables



The screenshot shows the RStudio IDE interface. The script editor on the left contains the following R code:

```
1 #####
2 ###
3 ### File: 01_data_visualisation.r
4 ### Purpose: Load required packages and functions used to
5 ### create interactive maps.
6 ### Authors: Gabriel Rodrigues Palma
7 ### Date: 12/06/25
8 ###
9 #####
10 # Load required libraries
11 source('00_source.r')
12
13 # Load main dataset
14 df <- read_excel("input_data/reviewStatistics.xlsm")
15 df
16
17 # What journals have been publishing the
18
19 # RQ1 Visualization 1: Evolution of ML approaches over time
```

The Environment pane on the right shows the following data objects:

Object	Details
df	54 obs. of 15 variables
df_ml_expanded	26 obs. of 3 variables

The console at the bottom shows the R startup message:

```
R 4.5.0 - ~/Desktop/Phd/Projects/SpringerBookChapters/Code/
Natural language support but running in an English locale

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

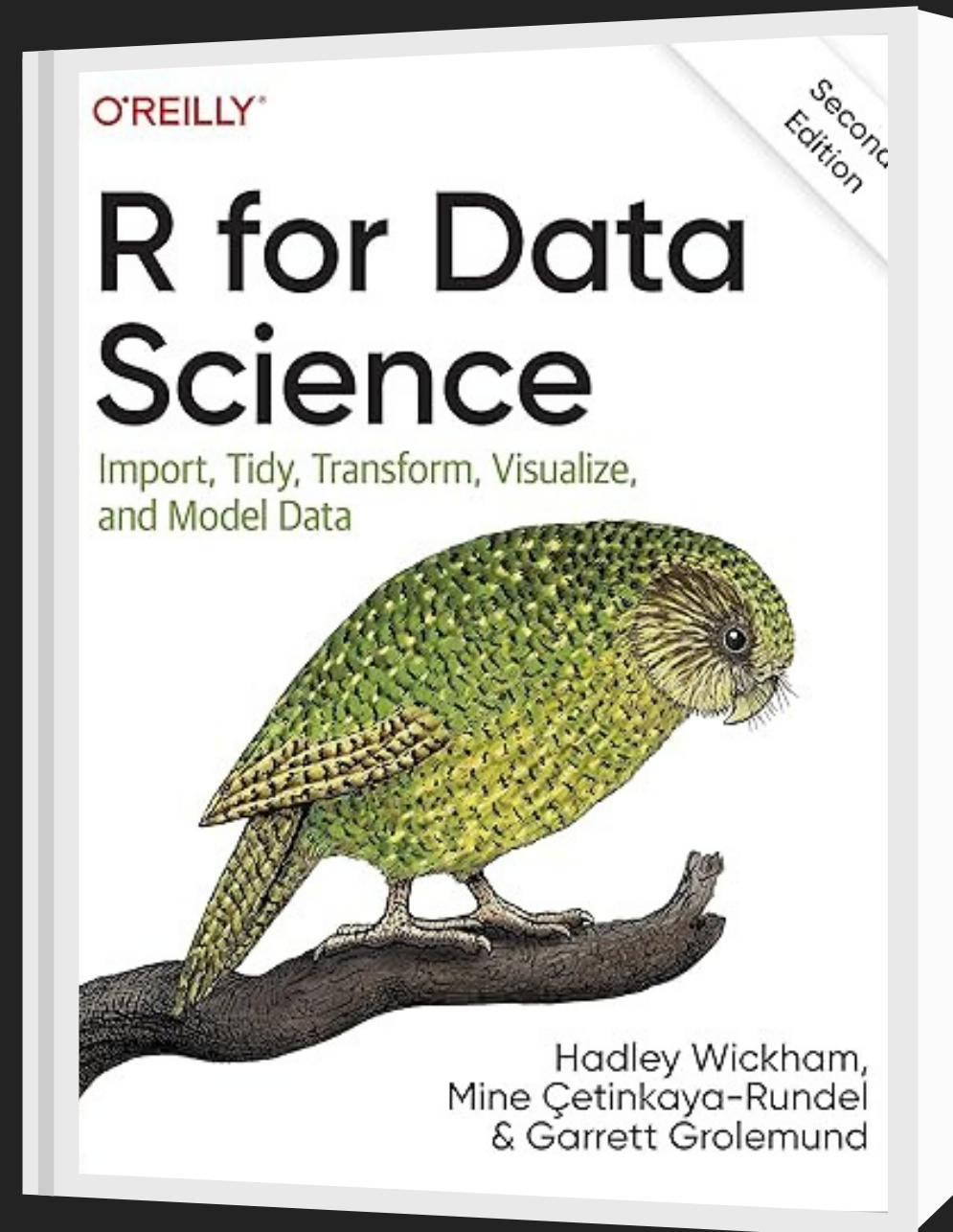
Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

[Workspace loaded from ~/Desktop/Phd/Projects/SpringerBookChapters/Code/.RData]
> |
```

Tidyverse for Ecologists
(TIDY01)



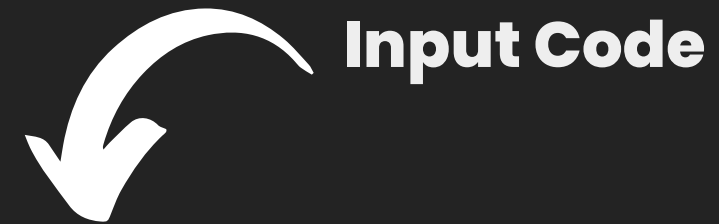
R Syntax & Variables



Tidyverse for Ecologists
(TIDY01)



R Syntax & Variables



R Input

```
# Creating data frames for ecological data
field_survey <- data.frame(
  plot_id = 1:4,
  species = c("Oak", "Pine", "Birch", "Maple"),
  dbh_cm = c(28.5, 22.1, 19.8, 31.2),
  height_m = c(12.3, 8.7, 7.2, 14.1)
)
print(field_survey)
```



Console Output

	plot_id	species	dbh_cm	height_m
1	1	Oak	28.5	12.3
2	2	Pine	22.1	8.7
3	3	Birch	19.8	7.2
4	4	Maple	31.2	14.1

R Syntax & Variables

Input Code

R Input

```
# Creating data frames for ecological data
field_survey <- data.frame(
  plot_id = 1:4,
  species = c("Oak", "Pine", "Birch", "Maple"),
  dbh_cm = c(28.5, 22.1, 19.8, 31.2),
  height_m = c(12.3, 8.7, 7.2, 14.1)
)
print(field_survey)
```



Console Output

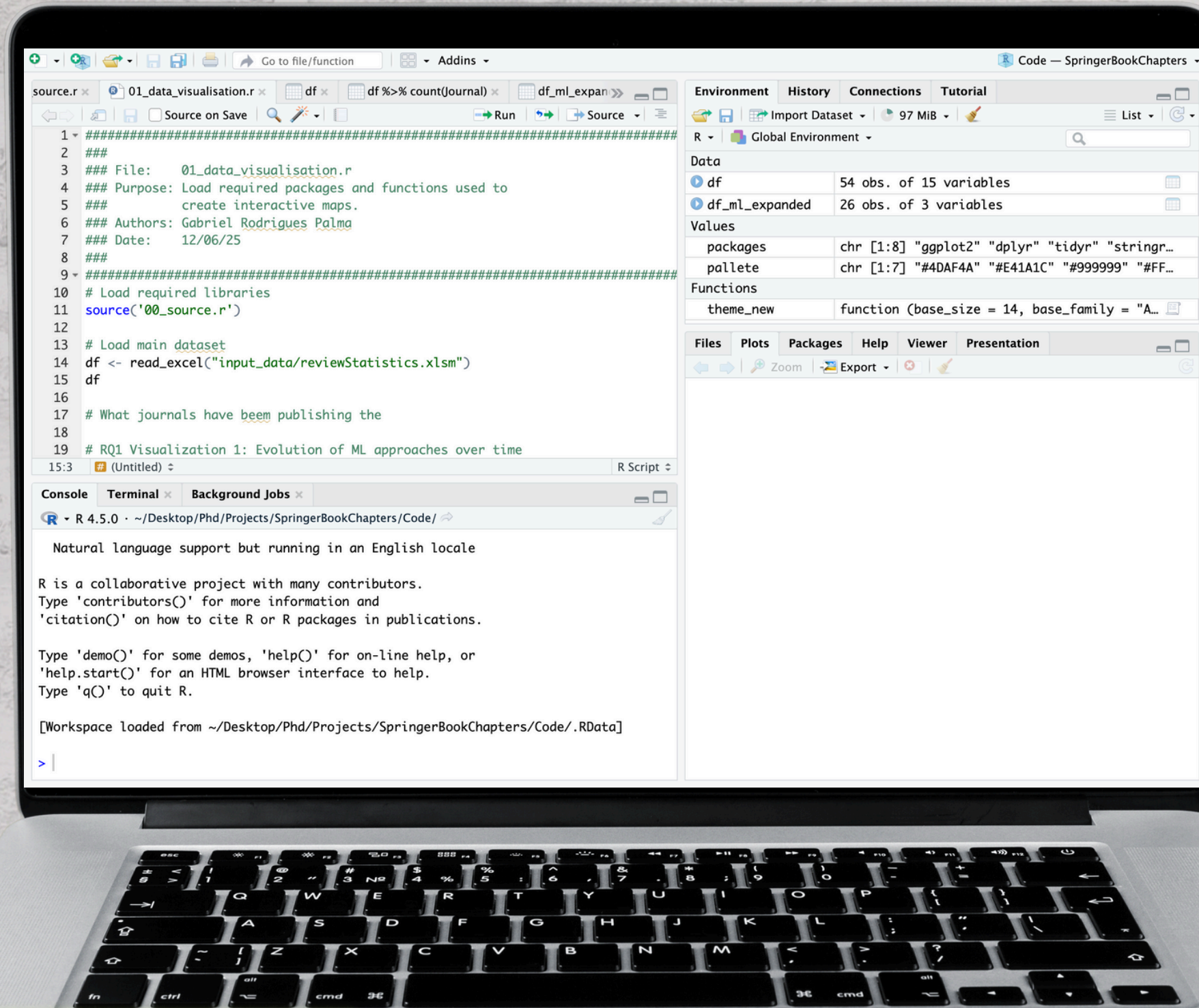
	plot_id	species	dbh_cm	height_m
1	1	Oak	28.5	12.3
2	2	Pine	22.1	8.7
3	3	Birch	19.8	7.2
4	4	Maple	31.2	14.1

Output

Tidyverse for Ecologists
(TIDY01)



R Syntax & Variables



Tidyverse for Ecologists
(TIDY01)



R Syntax & Variables

R Input

```
# Basic variable assignment for species data
species_name <- "Quercus robur"
tree_height <- 15.3
measurement_date <- "2024-06-15"
cat("Species:", species_name, "\nHeight:", tree_height, "meters\n")
cat("Date:", measurement_date, "\n")
```



Console Output

```
Species: Quercus robur
Height: 15.3 meters
Date: 2024-06-15
```



R Syntax & Variables

R Input

```
# Wildlife population density calculation
deer_count <- 47
survey_area <- 125.5
population_density <- deer_count / survey_area
cat("Population density:", population_density, "deer per hectare\n")
```



Console Output

```
Population density: 0.3745 deer per hectare
```



R Syntax & Variables

R Input

```
# Environmental measurements
temperature <- 18.5
humidity <- 72
wind_speed <- 3.2
weather_station <- "Alpine_Station_02"
print(paste("Weather at", weather_station, ":", temperature, "°C"))
```



Console Output

```
[1] "Weather at Alpine_Station_02 : 18.5 °C"
```



R Syntax & Variables

R Input

```
# Marine ecosystem variables
ocean_temperature <- 14.2
salinity <- 35.1
depth_meters <- 250
marine_protected_area <- TRUE
fish_abundance <- 156
cat("Ocean conditions:", ocean_temperature, "°C,", salinity, "ppt\n")
```



Console Output

```
Ocean conditions: 14.2 °C, 35.1 ppt
```



R Syntax & Variables

R Input

```
# Conservation status data
species_scientific <- "Panthera tigris"
common_name <- "Bengal Tiger"
population_estimate <- 2500
conservation_status <- "Endangered"
cat(common_name, "(", species_scientific, "):")
cat(population_estimate, "individuals\n")
```



Console Output

```
Bengal Tiger ( Panthera tigris ): 2500 individuals
```



R Syntax & Variables

R Input

```
site_info <- list(site="Delta", lat=44.25, protected=FALSE)
cat("Value:", site_info$site, site_info$lat, site_info$protected, "\n")
cat("Type:", typeof(site_info), "\n")
```



Console Output

```
Value: Delta 44.25 FALSE
Type: list
```

R Syntax & Variables

R Input

```
is_endangered <- TRUE  
cat("Value:", is_endangered, "\n")  
cat("Type:", typeof(is_endangered), "\n")
```



Console Output

```
Value: TRUE  
Type: logical
```



R Syntax & Variables

R Input

```
temperature_readings <- c(17.2, 18.5, 19.0)
cat("Value:", temperature_readings, "\n")
cat("Type:", typeof(temperature_readings), "\n")
```



Console Output

```
Value: 17.2 18.5 19
Type: double
```



R Syntax & Variables

R Input

```
population_count <- 250  
cat("Value:", population_count, "\n")  
cat("Type:", typeof(population_count), "\n")
```



Console Output

```
Value: 250  
Type: double
```



R Syntax & Variables

R Input

```
species_name <- "Canis lupus"  
cat("Value:", species_name, "\n")  
cat("Type:", typeof(species_name), "\n")
```



Console Output

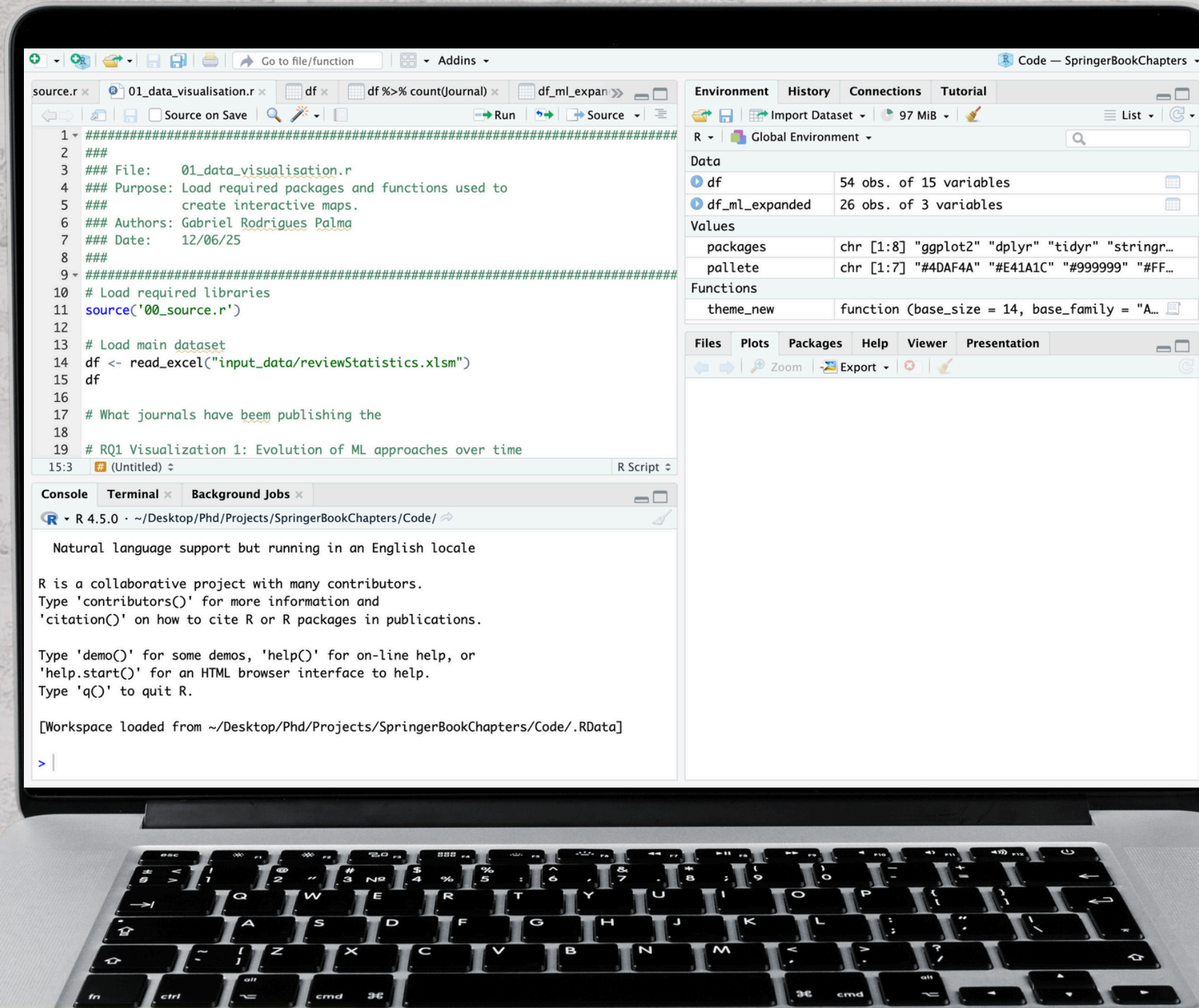
```
Value: Canis lupus  
Type: character
```



Tidyverse for Ecologists
(TIDY01)



R Syntax & Variables



The screenshot shows the RStudio IDE interface. The main editor window displays an R script file named '01_data_visualisation.r'. The script contains comments and code for loading packages and data. The Environment pane on the right shows the global environment with two data frames: 'df' (54 observations, 15 variables) and 'df_ml_expanded' (26 observations, 3 variables). The console at the bottom shows the R version (4.5.0) and the workspace loaded from a file named '.RData'.

```
1 #####
2 ###
3 ### File: 01_data_visualisation.r
4 ### Purpose: Load required packages and functions used to
5 ### create interactive maps.
6 ### Authors: Gabriel Rodrigues Palma
7 ### Date: 12/06/25
8 ###
9 #####
10 # Load required libraries
11 source('00_source.r')
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13 # Load main dataset
14 df <- read_excel("input_data/reviewStatistics.xlsm")
15 df
16
17 # What journals have been publishing the
18
19 # RQ1 Visualization 1: Evolution of ML approaches over time
```

Environment

Object	Class	Attributes
df	data.frame	54 obs. of 15 variables
df_ml_expanded	data.frame	26 obs. of 3 variables

Values

Object	Class	Attributes
packages	chr [1:8]	"ggplot2" "dplyr" "tidyr" "stringr..."
pallette	chr [1:7]	"#4DAF4A" "#E41A1C" "#999999" "#FF..."

Functions

Object	Class	Attributes
theme_new	function	(base_size = 14, base_family = "A..."

Console

```
R - R 4.5.0 - ~/Desktop/Phd/Projects/SpringerBookChapters/Code/
Natural language support but running in an English locale

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
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Type 'q()' to quit R.

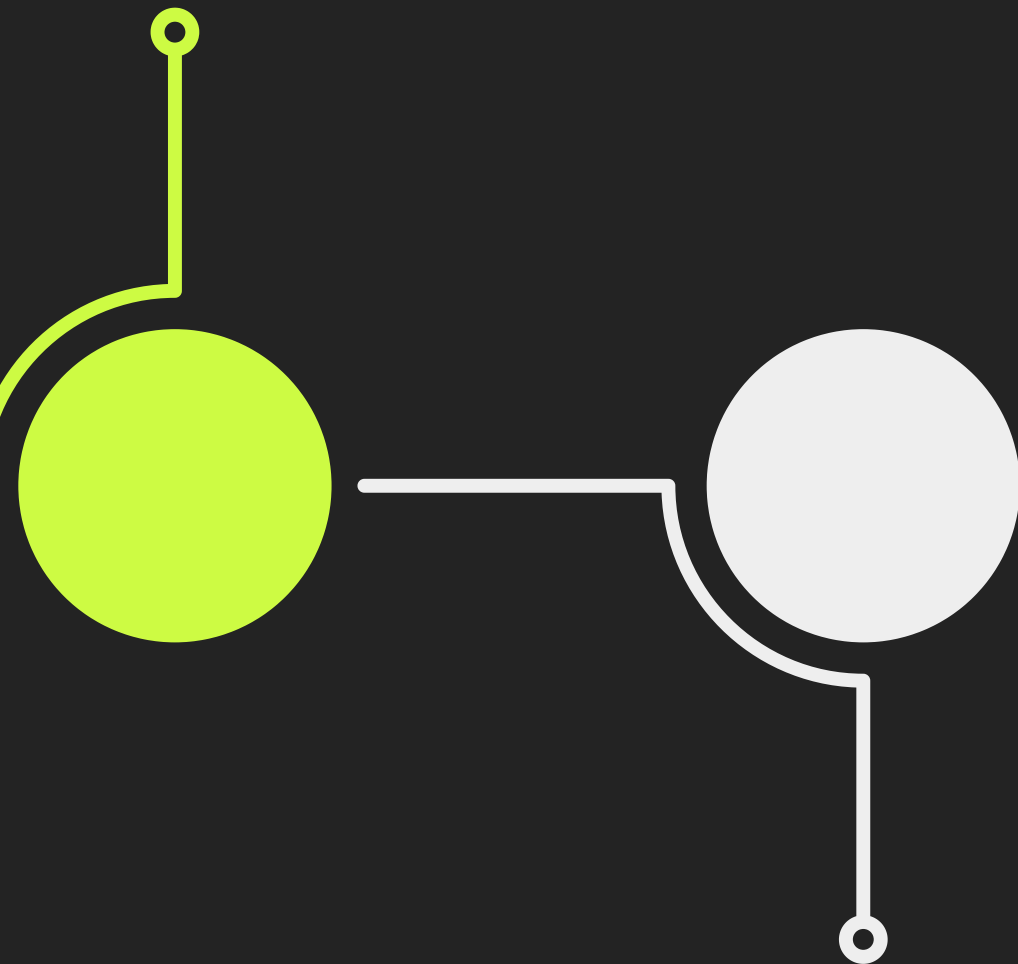
[Workspace loaded from ~/Desktop/Phd/Projects/SpringerBookChapters/Code/.RData]
> |
```

Tidyverse for Ecologists
(TIDY01)



Day 1 (9:30 – 17:30)

R Syntax & Variables
(9:35 – 11:05)



Data Types/Conditionals
(11:05 – 12:30)

Tidyverse for Ecologists
(TIDY01)



Data Types/Conditionals

R Input

```
# Species conservation status classification
species_population <- 250
if (species_population < 100) {
  status <- "Critically Endangered"
} else if (species_population < 500) {
  status <- "Endangered"
} else {
  status <- "Vulnerable"
}
cat("Conservation Status:", status, "\n")
```



Console Output

Conservation Status: Endangered



Data Types/Conditionals

R Input

```
# Habitat suitability assessment
temperature <- 22.5
precipitation <- 850
if (temperature > 20 && precipitation > 800) {
  habitat_quality <- "Optimal"
} else if (temperature > 15 && precipitation > 500) {
  habitat_quality <- "Suitable"
} else {
  habitat_quality <- "Marginal"
}
cat("Habitat Quality:", habitat_quality, "\n")
```



Console Output

```
Habitat Quality: Optimal
```



Data Types/Conditionals

R Input

```
# Water quality classification
ph_level <- 7.2
dissolved_oxygen <- 8.5
if (ph_level >= 6.5 && ph_level <= 8.5 && dissolved_oxygen >= 7) {
  water_quality <- "Excellent"
} else if (ph_level >= 6.0 && ph_level <= 9.0 && dissolved_oxygen >= 5) {
  water_quality <- "Good"
} else {
  water_quality <- "Poor"
}
cat("Water Quality:", water_quality, "\n")
```



Console Output

```
Water Quality: Excellent
```


Data Types/Conditionals

R Input

```
# Species distribution modeling
elevation <- 1500
slope <- 25
aspect <- "North"
if (elevation > 1000 && elevation < 2000 && slope < 30) {
  if (aspect == "North" || aspect == "Northeast") {
    habitat_suitability <- "High"
  } else {
    habitat_suitability <- "Medium"
  }
} else {
  habitat_suitability <- "Low"
}
cat("Habitat Suitability:", habitat_suitability, "\n")
```



Console Output

```
Habitat Suitability: High
```

Data Types/Conditionals

R Input

```
initial_pop <- 150
growth_rate <- 0.08
carrying_capacity <- 1000
time_steps <- seq(0, 10, by = 0.5)
logistic_growth <- initial_pop / (1 + ((carrying_capacity - initial_pop) /
  initial_pop) * exp(-growth_rate * time_steps))
beta_function <- beta(3, 5)
gamma_function <- gamma(4.5)
combinations <- choose(10, 3)
factorial_calc <- factorial(6)
bessel_func <- besselJ(2.5, 1)
cat("Pop at t=5:", round(logistic_growth[11], 2), "\n")
cat("Beta function:", round(beta_function, 4), "\n")
cat("Combinations:", combinations, "\n")
```



Console Output

```
Pop at t=5: 223.13
Beta function: 0.0083
Combinations: 120
```


Data Types/Conditionals

R Input

```
community_matrix <- matrix(c(12, 8, 15, 6, 20, 4, 18, 9, 14, 7, 11, 3),  
                             nrow = 3, ncol = 4)  
matrix_transpose <- t(community_matrix)  
row_sums <- rowSums(community_matrix)  
col_means <- colMeans(community_matrix)  
matrix_multiplication <- community_matrix %*% t(community_matrix)  
determinant_val <- det(matrix_multiplication)  
eigen_values <- eigen(matrix_multiplication)$values  
matrix_inverse <- solve(matrix_multiplication)  
cat("Row sums:", row_sums, "\n")  
cat("Column means:", col_means, "\n")  
cat("Determinant:", round(determinant_val, 2), "\n")
```



Console Output

```
Row sums: 41 51 35  
Column means: 14.66667 8.00000 15.66667 6.00000  
Determinant: 171428.44
```



Data Types/Conditionals

R Input

```
species_counts <- c(45, 32, 28, 67, 23, 51, 39, 44)
mean_abundance <- mean(species_counts)
median_abundance <- median(species_counts)
standard_deviation <- sd(species_counts)
variance_value <- var(species_counts)
min_count <- min(species_counts)
max_count <- max(species_counts)
sum_total <- sum(species_counts)
range_values <- range(species_counts)
quantiles <- quantile(species_counts, c(0.25, 0.75))
cat("Mean:", mean_abundance, "\n")
cat("SD:", round(standard_deviation, 2), "\n")
cat("Range:", min_count, "to", max_count, "\n")
```



Console Output

```
Mean: 41.125
SD: 14.36
Range: 23 to 67
```


Data Types/Conditionals

R Input

```
population_size <- 2500
square_root_pop <- sqrt(population_size)
natural_log <- log(population_size)
log_base_10 <- log10(population_size)
exponential_growth <- exp(2.5)
sine_angle <- sin(pi/4)
cosine_angle <- cos(pi/3)
tangent_angle <- tan(pi/6)
absolute_value <- abs(-15.7)
ceiling_value <- ceiling(12.3)
floor_value <- floor(12.8)
cat("Square root:", round(square_root_pop, 2), "\n")
cat("Natural log:", round(natural_log, 2), "\n")
cat("Exponential:", round(exponential_growth, 2), "\n")
```



Console Output

```
Square root: 50
Natural log: 7.82
Exponential: 12.18
```

Data Types/Conditionals

R Input

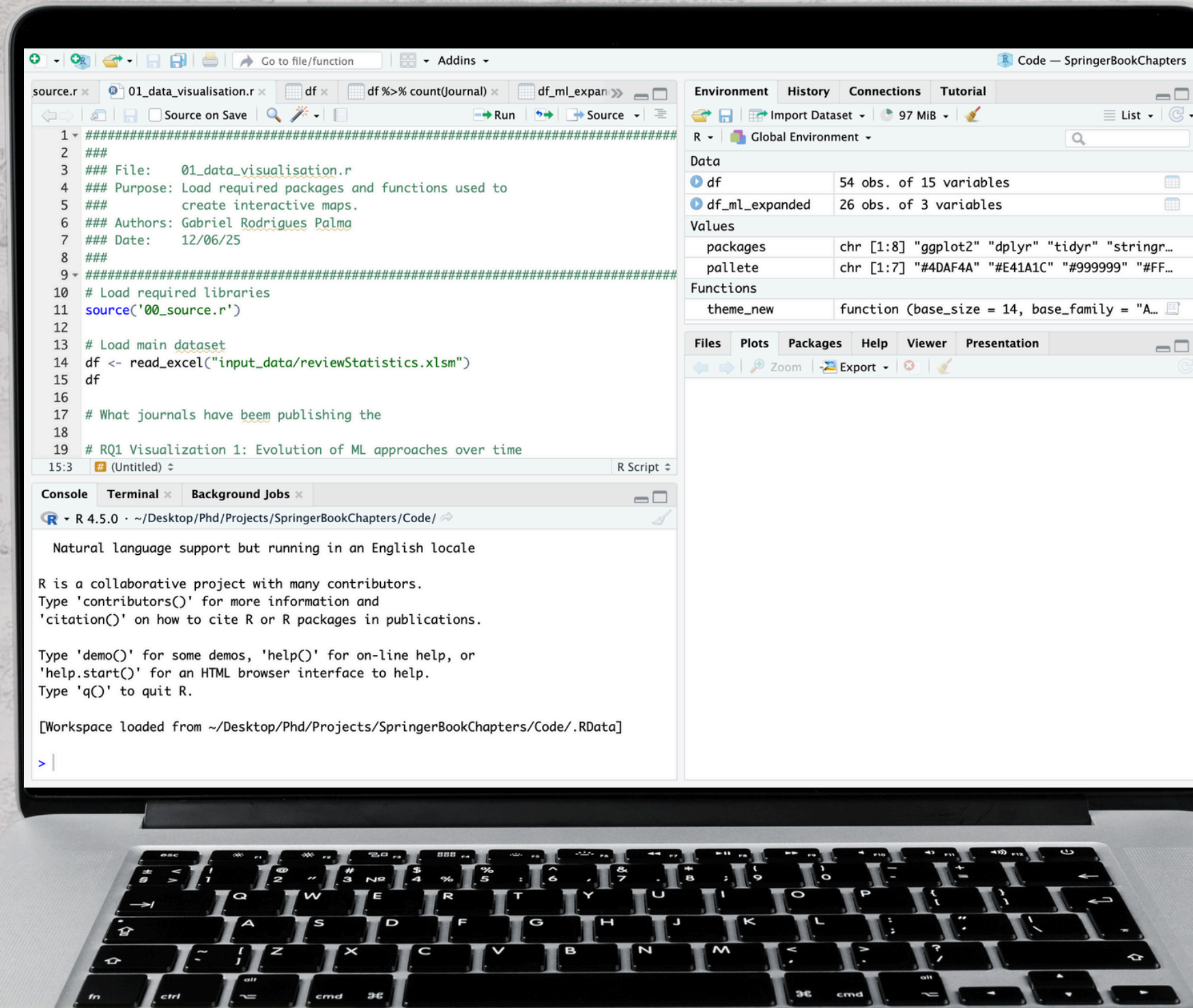
```
tree_diameter <- 25.3
tree_height <- 18.7
basic_addition <- tree_diameter + tree_height
basic_subtraction <- tree_height - tree_diameter
basic_multiplication <- tree_diameter * 2.5
basic_division <- tree_height / tree_diameter
exponentiation <- tree_diameter ^ 2
modulo_operation <- 127 %% 5
integer_division <- 127 %/% 5
cat("Addition:", basic_addition, "\n")
cat("Multiplication:", basic_multiplication, "\n")
cat("Power:", exponentiation, "\n")
cat("Modulo:", modulo_operation, "\n")
```



Console Output

```
Addition: 44
Multiplication: 63.25
Power: 640.09
Modulo: 2
```


Data Types/Conditionals



The screenshot displays the RStudio interface. The main editor shows an R script with the following content:

```
1 #####
2 ###
3 ### File: 01_data_visualisation.r
4 ### Purpose: Load required packages and functions used to
5 ### create interactive maps.
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15 df
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17 # What journals have been publishing the
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19 # RQ1 Visualization 1: Evolution of ML approaches over time
```

The console at the bottom shows the R session output:

```
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Type 'q()' to quit R.

[Workspace loaded from ~/Desktop/Phd/Projects/SpringerBookChapters/Code/.RData]
> |
```

The Environment pane on the right shows the following data objects:

Object	Details
df	54 obs. of 15 variables
df_ml_expanded	26 obs. of 3 variables

The Values pane shows the following data:

Variable	Values
packages	chr [1:8] "ggplot2" "dplyr" "tidyr" "stringr..."
pallette	chr [1:7] "#4DAF4A" "#E41A1C" "#999999" "#FF..."

The Functions pane shows the following function:

Function	Details
theme_new	function (base_size = 14, base_family = "A..."

Tidyverse for Ecologists
(TIDY01)



Lunch time



Tidyverse for Ecologists
(TIDY01)

 stats

